

# MLB - Player Value Analysis

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2025-02-06

## Scrape Player Salary Data

This function scrapes MLB player data from *spotrac.com* for a given year using the *rvest* package.

```
scrape_salaries <- function(year) {  
  url <- paste0("https://www.spotrac.com/mlb/rankings/player/_/year/", year, "/sort/cap_total")  
  webpage <- read_html(url)  
  
  Name <- html_nodes(webpage, ".col-md-8 .link") %>%  
    html_text() %>%  
    sapply(function(x) gsub("\n", "", x)) %>%  
    trimws()  
  
  Salary <- html_nodes(webpage, ".col-md-8 .medium") %>%  
    html_text() %>%  
    sapply(function(x) gsub("\n", "", x)) %>%  
    trimws()  
  
  data.frame(Name, Salary)  
}
```

## Scrape Player Statistics

This function scrapes MLB player statistics from *espn.com* for a given stat type and year.

```
scrape_stats <- function(stat_type, year) {  
  url <- paste0("https://www.espn.com/mlb/stats/player/_/view/batting/season/", year, "/seasontype/2/tal")  
  webpage <- read_html(url)  
  
  Stat <- html_nodes(webpage, ".Table__TD:nth-child(9)") %>%  
    html_text()  
  
  Name <- html_nodes(webpage, ".mr7 .AnchorLink") %>%  
    html_text()  
  
  data.frame(Name, Stat)  
}
```

## Cleaning Data

These functions clean salary data by converting them into numeric values.

```

clean_salary_data <- function(data) {
  data$Salary <- as.numeric(gsub("\\$", "", gsub(",", "", gsub(" ", "", data$Salary))))
  data
}

clean_stat_data <- function(data, stat_name) {
  data[[stat_name]] <- as.numeric(gsub("\\$", "", gsub(",", "", gsub(" ", "", data[[stat_name]]))))
  data
}

```

## Merging Data

This function merges salary data with specific statistics based on player names.

```

merge_data <- function(salary_data, stat_data, stat_name) {
  colnames(stat_data) <- c("Name", stat_name)
  merged_data <- merge(salary_data, stat_data, by = "Name", all = FALSE)
  clean_stat_data(merged_data, stat_name)
}

```

## Executing Scraping Functions

This function collects and merges salary and performance statistics for a given year.

```

perform_analysis <- function(year) {
  # Scrape data
  salary_data <- scrape_salaries(year)
  hr_data <- scrape_stats("homeRuns", year)
  rbi_data <- scrape_stats("RBIs", year)
  hits_data <- scrape_stats("hits", year)
  sb_data <- scrape_stats("stolenBases", year)
  bb_data <- scrape_stats("walks", year)
  runs_data <- scrape_stats("runs", year)

  # Clean salary data
  salary_data <- clean_salary_data(salary_data)

  # Merge data
  merged_hr <- merge_data(salary_data, hr_data, "HR")
  merged_rbi <- merge_data(salary_data, rbi_data, "RBI")
  merged_hits <- merge_data(salary_data, hits_data, "Hits")
  merged_sb <- merge_data(salary_data, sb_data, "SB")
  merged_bb <- merge_data(salary_data, bb_data, "BB")
  merged_runs <- merge_data(salary_data, runs_data, "Runs")

  # Return the merged data
  list(
    HR = merged_hr,
    RBI = merged_rbi,
    Hits = merged_hits,
    SB = merged_sb,

```

```

    BB = merged_bb,
    Runs = merged_runs
  )
}

```

## Analysis by Year

Running perform\_analysis function for each year from 2012-2024.

```

years <- 2012:2024
results_by_year <- list()
for (year in years) {
  results_by_year[[as.character(year)]] <- perform_analysis(year)
}

```

## Calculating Averages

This function computes the average salary and stats per year.

```

calculate_means <- function(year) {
  dataset <- get(paste0("MLBresults_", year)) # Dynamically get dataset

  stats <- c("HR", "RBI", "Runs", "BB", "SB", "Hits") # Relevant stat categories

  means <- setNames(lapply(stats, function(stat) {
    list(
      mean(dataset[[stat]][[stat]], na.rm = TRUE),
      mean(dataset[[stat]]$Salary, na.rm = TRUE)
    )
  }), stats)

  # Flatten and rename list elements
  means <- unlist(means, recursive = FALSE)
  names(means) <- c(rbind(paste0(stats, "_", stats), paste0(stats, "_Salary")))

  return(means)
}

```

## Calculating Ratios for Player Value

This function calculates the salary-per-stat ratios for each year.

```

calculate_ratios <- function(year) {
  means <- results_by_year[[as.character(year)]]
  return(list(
    value_per_HR = means[["HR_Salary"]] / means[["HR_HR"]],
    value_per_RBI = means[["RBI_Salary"]] / means[["RBI_RBI"]],
    value_per_Runs = means[["Runs_Salary"]] / means[["Runs_Runs"]],
    value_per_BB = means[["BB_Salary"]] / means[["BB_BB"]],
    value_per_SB = means[["SB_Salary"]] / means[["SB_SB"]],
  ))
}

```

```

    value_per_Hits = means[["Hits_Salary"]] / means[["Hits_Hits"]]
  })
}

```

## Calculating a Player's Value

This function estimates a player's value based on their stats in a given year.

```

calculate_player_value <- function(HR, Runs, RBI, BB, Hits, SB, Year) {
  if (Year < 2012 || Year > 2024) stop("Year must be between 2012 and 2024")
  value_of_player <- (
    (ratios_by_year[[as.character(Year)]]["value_per_HR"]] * HR / 6) +
    (ratios_by_year[[as.character(Year)]]["value_per_RBI"]] * RBI / 6) +
    (ratios_by_year[[as.character(Year)]]["value_per_Runs"]] * Runs / 6) +
    (ratios_by_year[[as.character(Year)]]["value_per_BB"]] * BB / 6) +
    (ratios_by_year[[as.character(Year)]]["value_per_SB"]] * SB / 6) +
    (ratios_by_year[[as.character(Year)]]["value_per_Hits"]] * Hits / 6)
  )
  return(value_of_player)
}

```

## Example Usage

```
HR <- as.numeric(readline("How many homeruns did the player have? "))
```

```
## How many homeruns did the player have?
```

```
Runs <- as.numeric(readline("How many runs did the player have? "))
```

```
## How many runs did the player have?
```

```
RBI <- as.numeric(readline("How many RBIs did the player have? "))
```

```
## How many RBIs did the player have?
```

```
BB <- as.numeric(readline("How many walks did the player have? "))
```

```
## How many walks did the player have?
```

```
Hits <- as.numeric(readline("How many hits did the player have? "))
```

```
## How many hits did the player have?
```

```
SB <- as.numeric(readline("How many stolen bases did the player have? "))
```

```
## How many stolen bases did the player have?
```

```
Year <- as.numeric(readline("What year did they have these stats? "))
```

```
## What year did they have these stats?
```

```
player_value <- calculate_player_value(HR, Runs, RBI, BB, Hits, SB, Year)
```

```
## Error in if (Year < 2012 || Year > 2024) stop("Year must be between 2012 and 2024"): missing value where TRUE/FALSE needed
```

```
cat("The value of the player for the year", Year, "is:", player_value, "\n")
```

```
## Error: object 'player_value' not found
```